Lecture Notes: Tensor Calculus

Based on Youtube series by eigenchris

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1 Multi-variable Calculus Review

For this lecture notes on Tensor Calculus, there are some concepts and notes that we need to quickly review on multi-variable calculus, which need to be clear and understood before proceeding.
Here we'll give a brief review of the following topics:

- Single-variable Derivatives and chain rule;
- Single-variable Integrals and differentials (change of variable);
- Multi-variable Partial derivatives and chain rule;
- Gradient of a function;
- Directional Derivatives
- Multi-variable Integrals and differentials;
- Arch length, velocity vector tangent to a curve.

1.1 Single-Variable Derivatives and Chain Rule

The derivative of a function $f(x)$ with respect to x measures the instantaneous rate of change:

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

Basic derivative rules:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}(x^n) &= nx^{n-1} \\
\frac{\mathrm{d}}{\mathrm{d}x}(e^x) &= e^x \\
\frac{\mathrm{d}}{\mathrm{d}x}(\sin x) &= \cos x \\
\frac{\mathrm{d}}{\mathrm{d}x}(\cos x) &= -\sin x \\
\frac{\mathrm{d}}{\mathrm{d}x}(\ln x) &= \frac{1}{x}
\end{aligned}$$

Sum and product rules:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}(f+g) &= \frac{\mathrm{d}f}{\mathrm{d}x} + \frac{\mathrm{d}g}{\mathrm{d}x} \\
\frac{\mathrm{d}}{\mathrm{d}x}(fg) &= f\frac{\mathrm{d}g}{\mathrm{d}x} + g\frac{\mathrm{d}f}{\mathrm{d}x}
\end{aligned}$$

Chain rule: For a composition of functions $f(g(x))$:

$$\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = \frac{\mathrm{d}f}{\mathrm{d}g} \cdot \frac{\mathrm{d}g}{\mathrm{d}x}$$

More explicitly, if $y = f(u)$ and $u = g(x)$, then:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u} \cdot \frac{\mathrm{d}u}{\mathrm{d}x}$$

Example: Let $y = \sin(x^2)$. Setting $u = x^2$, we have $y = \sin(u)$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \cos(u) \cdot 2x = 2x\cos\left(x^2\right)$$

1.2 Single-Variable Integrals and Change of Variable

The definite integral of $f(x)$ from a to b represents the signed area under the curve:

$$\int_a^b f(x)\,dx$$

Fundamental Theorem of Calculus: If $F(x)$ is an antiderivative of $f(x)$ (i.e., $\frac{\mathrm{d}F}{\mathrm{d}x} = f$), then:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Basic integrals:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int e^x\,dx = e^x + C$$

$$\int \sin x\,dx = -\cos x + C$$

$$\int \cos x\,dx = \sin x + C$$

Change of Variable (Substitution): To integrate $f(g(x))g'(x)$, substitute $u = g(x)$, so that $du = g'(x)\,dx$:

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$

For definite integrals, the limits change accordingly: if $u = g(x)$, then $x = a \to u = g(a)$ and $x = b \to u = g(b)$.

Example: Evaluate $\int x\cos(x^2)\,dx$. Let $u = x^2$, then $du = 2x\,dx$, so $x\,dx = \frac{1}{2}du$:

$$\int x\cos\left(x^2\right)dx = \int \cos(u) \cdot \frac{1}{2}\,du = \frac{1}{2}\sin(u) + C = \frac{1}{2}\sin\left(x^2\right) + C$$

1.3 Multi-Variable Partial Derivatives and Chain Rule

For a function $f(x, y)$ of multiple variables, the **partial derivative** with respect to one variable treats all other variables as constants.

Notation:

$$\frac{\partial f}{\partial x}, \quad \frac{\partial f}{\partial y}, \quad f_x, \quad f_y$$

Definition:

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$

Example: For $f(x, y) = x^2 y + \sin(xy)$:

$$\frac{\partial f}{\partial x} = 2xy + y\cos(xy)$$
$$\frac{\partial f}{\partial y} = x^2 + x\cos(xy)$$

Multivariable Chain Rule: Consider $z = f(x, y)$ where $x = x(t)$ and $y = y(t)$ are functions of t. Then:

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}$$

More generally, if $z = f(x, y)$ and $x = x(u, v)$, $y = y(u, v)$, then:

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u}$$

Example: Let $z = x^2 + y^2$ with $x = r\cos\theta$ and $y = r\sin\theta$. Then:

$$\begin{aligned}\frac{\partial z}{\partial r} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r} \\ &= (2x)(\cos\theta) + (2y)(\sin\theta) \\ &= 2r\cos^2\theta + 2r\sin^2\theta = 2r\end{aligned}$$

Einstein Summation Convention: Using indexed notation where x^i represents the i-th coordinate, the chain rule can be written compactly as:

$$\frac{\partial f}{\partial u^j} = \frac{\partial f}{\partial x^i}\frac{\partial x^i}{\partial u^j}$$

where summation over the repeated index i is implied (no explicit $\sum$ symbol needed). This convention will be used extensively in tensor calculus.

1.4 Gradient of a Function

The **gradient** of a scalar function $f(x, y, z)$ is a vector field that points in the direction of greatest rate of increase of the function.

Definition: For a function $f : \mathbb{R}^n \to \mathbb{R}$, the gradient is:

$$\boldsymbol{\nabla} f = \nabla f = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$$

In three dimensions:

$$\boldsymbol{\nabla} f = \frac{\partial f}{\partial x}\hat{\imath} + \frac{\partial f}{\partial y}\hat{\jmath} + \frac{\partial f}{\partial z}\hat{k}$$

Geometric interpretation:

- $\boldsymbol{\nabla} f$ points in the direction of steepest ascent of f
- $|\boldsymbol{\nabla} f|$ gives the rate of change in that direction
- $\boldsymbol{\nabla} f$ is perpendicular to level curves (2D) or level surfaces (3D) of f

Example: For $f(x, y) = x^2 + y^2$:

$$\boldsymbol{\nabla} f = 2x\hat{\imath} + 2y\hat{\jmath} = 2(x\hat{\imath} + y\hat{\jmath})$$

At point $(1, 1)$: $\boldsymbol{\nabla} f = 2\hat{\imath} + 2\hat{\jmath}$, pointing radially outward.

Applications:

- *Optimization*: Finding maxima/minima (critical points where $\boldsymbol{\nabla} f = 0$)
- *Physics*: Conservative forces $\vec{F} = -\boldsymbol{\nabla} U$ (gravitational, electric fields)
- *Machine learning*: Gradient descent for minimizing loss functions
- *Fluid dynamics*: Pressure gradients drive fluid flow

1.5 Directional Derivatives

While partial derivatives measure the rate of change along coordinate axes, the **directional derivative** measures how fast a function changes in an arbitrary direction.

Definition: The directional derivative of f at point $\vec{a}$ in the direction of unit vector $\vec{u}$ is:

$$D_{\vec{u}} f(\vec{a}) = \lim_{h \to 0} \frac{f(\vec{a} + h\vec{u}) - f(\vec{a})}{h}$$

This gives the instantaneous rate of change of f as we move from $\vec{a}$ in direction $\vec{u}$.

Key requirement: $\vec{u}$ must be a *unit vector* (i.e., $|\vec{u}| = 1$) for the directional derivative to represent the true rate of change per unit distance.

Formula using gradient: The directional derivative can be computed using the dot product:

$$D_{\vec{u}} f = \boldsymbol{\nabla} f \cdot \vec{u} = |\boldsymbol{\nabla} f||\vec{u}| \cos\theta = |\boldsymbol{\nabla} f| \cos\theta$$

where θ is the angle between $\boldsymbol{\nabla} f$ and $\vec{u}$.

Important properties:

- **Maximum rate:** $D_{\vec{u}}f$ is maximized when $\vec{u}$ points in the direction of $\boldsymbol{\nabla} f$ (i.e., $\theta = 0$):

$$\max_{\vec{u}} D_{\vec{u}}f = |\boldsymbol{\nabla} f|$$

- **Minimum rate:** $D_{\vec{u}}f$ is minimized when $\vec{u}$ points opposite to $\boldsymbol{\nabla} f$ (i.e., $\theta = \pi$):

$$\min_{\vec{u}} D_{\vec{u}}f = -|\boldsymbol{\nabla} f|$$

- **Zero rate:** $D_{\vec{u}}f = 0$ when $\vec{u}$ is perpendicular to $\boldsymbol{\nabla} f$ (i.e., $\theta = \pi/2$), meaning along level curves/surfaces

Example 1: Find the directional derivative of $f(x, y) = x^2 + xy$ at $(1, 2)$ in the direction toward $(4, 6)$.

First, compute the gradient:

$$\boldsymbol{\nabla} f = (2x + y)\hat{\imath} + x\hat{\jmath} \quad \Rightarrow \quad \boldsymbol{\nabla} f(1, 2) = 4\hat{\imath} + \hat{\jmath}$$

The direction vector from $(1, 2)$ to $(4, 6)$ is $\vec{v} = 3\hat{\imath} + 4\hat{\jmath}$. Normalize it:

$$\vec{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{3\hat{\imath} + 4\hat{\jmath}}{\sqrt{9 + 16}} = \frac{3}{5}\hat{\imath} + \frac{4}{5}\hat{\jmath}$$

The directional derivative is:

$$D_{\vec{u}}f = \boldsymbol{\nabla} f \cdot \vec{u} = (4\hat{\imath} + \hat{\jmath}) \cdot \left(\frac{3}{5}\hat{\imath} + \frac{4}{5}\hat{\jmath}\right) = \frac{12}{5} + \frac{4}{5} = \frac{16}{5}$$

Example 2: For $f(x, y) = x^2 + y^2$ at $(3, 4)$, find the direction of maximum increase and the maximum rate.

$$\boldsymbol{\nabla} f = 2x\hat{\imath} + 2y\hat{\jmath} \quad \Rightarrow \quad \boldsymbol{\nabla} f(3, 4) = 6\hat{\imath} + 8\hat{\jmath}$$

The direction of maximum increase is along $\boldsymbol{\nabla} f$. The unit vector is:

$$\vec{u}_{\max} = \frac{6\hat{\imath} + 8\hat{\jmath}}{\sqrt{36 + 64}} = \frac{6\hat{\imath} + 8\hat{\jmath}}{10} = \frac{3}{5}\hat{\imath} + \frac{4}{5}\hat{\jmath}$$

The maximum rate of increase is:

$$|\boldsymbol{\nabla} f(3, 4)| = \sqrt{36 + 64} = 10$$

1.6 Multi-Variable Integrals and Differentials

Double and triple integrals extend the concept of integration to functions of multiple variables, computing volumes under surfaces or higher-dimensional analogues.

Double integral: For a function $f(x, y)$ over region R:

$$\iint_R f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx$$

where $dA = dx\,dy$ is the area element.

Triple integral: For a function $f(x, y, z)$ over region V:

$$\iiint_V f(x, y, z)\,dV$$

where $dV = dx\,dy\,dz$ is the volume element.

Change of Variables (Jacobian): When transforming coordinates $(x, y) \to (u, v)$, the area element transforms according to:

$$dx\,dy = |J|\,du\,dv$$

where J is the **Jacobian determinant**:

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

For three dimensions $(x, y, z) \to (u, v, w)$:

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

Example - Polar coordinates: For $x = r\cos\theta$, $y = r\sin\theta$:

$$J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r$$

Therefore: $dx\,dy = r\,dr\,d\theta$

This explains why integrals in polar coordinates have the extra factor of r:

$$\iint_R f(x, y)\,dx\,dy = \iint_{R'} f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta$$

Differential forms: A differential df of a function $f(x, y)$ is:

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$$

This represents the infinitesimal change in f due to infinitesimal changes dx and dy in the coordinates. More generally:

$$df = \boldsymbol{\nabla} f \cdot d\vec{r}$$

where $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$ is the displacement vector.

Einstein notation: Using indexed coordinates x^i, the differential can be written compactly as:

$$df = \frac{\partial f}{\partial x^i}dx^i$$

where summation over repeated index i is implied. This notation emphasizes that the differential is a linear combination of coordinate differentials weighted by partial derivatives.

1.7 Arc Length and Velocity Vector Tangent to a Curve

A **parametric curve** in space is described by a position vector $\vec{r}(t)$ that depends on parameter t:

$$\vec{r}(t) = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k}$$

Velocity vector: The derivative of position with respect to the parameter gives the velocity:

$$\vec{v}(t) = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \frac{\mathrm{d}x}{\mathrm{d}t}\hat{\imath} + \frac{\mathrm{d}y}{\mathrm{d}t}\hat{\jmath} + \frac{\mathrm{d}z}{\mathrm{d}t}\hat{k} = \dot{x}\hat{\imath} + \dot{y}\hat{\jmath} + \dot{z}\hat{k}$$

The velocity vector $\vec{v}(t)$ is **tangent to the curve** at each point and points in the direction of motion.

Speed: The magnitude of the velocity vector gives the speed:

$$|\vec{v}(t)| = \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}z}{\mathrm{d}t}\right)^2}$$

Unit tangent vector: Normalizing the velocity gives a unit vector tangent to the curve:

$$\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$$

Arc length: The length of a curve from $t = a$ to $t = b$ is:

$$s = \int_a^b |\vec{v}(t)|\, dt = \int_a^b \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}z}{\mathrm{d}t}\right)^2}\, dt$$

In differential form:

$$ds = |\vec{v}|\, dt = \sqrt{dx^2 + dy^2 + dz^2}$$

Arc length parameter: We can reparametrize a curve using arc length s as the parameter. Then:

$$\frac{\mathrm{d}\vec{r}}{\mathrm{d}s} = \vec{T}$$

is automatically a unit tangent vector (since $\left|\frac{\mathrm{d}\vec{r}}{\mathrm{d}s}\right| = 1$).

Example: Consider the helix $\vec{r}(t) = \cos(t)\hat{\imath} + \sin(t)\hat{\jmath} + t\hat{k}$ for $0 \le t \le 2\pi$.

Velocity vector:

$$\vec{v}(t) = -\sin(t)\hat{\imath} + \cos(t)\hat{\jmath} + \hat{k}$$

Speed:

$$|\vec{v}(t)| = \sqrt{\sin^2(t) + \cos^2(t) + 1} = \sqrt{2}$$

Arc length from $t = 0$ to $t = 2\pi$:

$$s = \int_0^{2\pi} \sqrt{2}\, dt = 2\pi\sqrt{2}$$

Unit tangent vector:

$$\vec{T}(t) = \frac{1}{\sqrt{2}}\left(-\sin(t)\hat{\imath} + \cos(t)\hat{\jmath} + \hat{k}\right)$$

Connection to tensor calculus: In general curvilinear coordinates, the tangent vectors to coordinate curves form the basis vectors of the coordinate system. Understanding how tangent vectors transform is fundamental to tensor analysis.

2 Basis vectors as partial derivatives and Jacobian

In two-dimensional Euclidean space, a point is described by Cartesian coordinates

$$(x, y)$$

The position vector is

$$\mathbf{R} = x\vec{e_x} + y\vec{e_y}$$

where the basis vectors are

$$\vec{e_x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \vec{e_y} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

These satisfy:

$$\vec{e_x} \cdot \vec{e_x} = 1, \quad \vec{e_y} \cdot \vec{e_y} = 1, \quad \vec{e_x} \cdot \vec{e_y} = 0.$$

Thus, $\{\vec{e_x}, \vec{e_y}\}$ is an orthonormal basis.
A key feature:

$$\frac{\partial \vec{e_x}}{\partial x} = \frac{\partial \vec{e_x}}{\partial y} = \frac{\partial \vec{e_y}}{\partial x} = \frac{\partial \vec{e_y}}{\partial y} = 0.$$

The basis vectors are constant everywhere in space.

We now define polar coordinates (r, θ) by the transformation

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

The position vector becomes

$$\mathbf{R} = r\cos\theta\,\vec{e_x} + r\sin\theta\,\vec{e_y}$$

We define the radial unit vector as

$$\tilde{\vec{e_r}} = \cos\theta\,\vec{e_x} + \sin\theta\,\vec{e_y}.$$

This points directly away from the origin.
And then define the angular unit vector as

$$\tilde{\vec{e_\theta}} = -\sin\theta\,\vec{e_x} + \cos\theta\,\vec{e_y}$$

This points in the direction of increasing θ.
These vectors satisfy

$$\tilde{\vec{e_r}} \cdot \tilde{\vec{e_r}} = 1, \quad \tilde{\vec{e_\theta}} \cdot \tilde{\vec{e_\theta}} = 1, \quad \tilde{\vec{e_r}} \cdot \tilde{\vec{e_\theta}} = 0$$

So they form an orthonormal basis at each point, but polar basis vectors depend on θ

Compute derivatives:

$$\frac{\partial \tilde{\vec{e_r}}}{\partial \theta} = -\sin\theta\, \vec{e_x} + \cos\theta\, \vec{e_y} = \tilde{\vec{e_\theta}}$$

$$\frac{\partial \tilde{\vec{e_\theta}}}{\partial \theta} = -\cos\theta\, \vec{e_x} - \sin\theta\, \vec{e_y} = -\tilde{\vec{e_r}}$$

Thus,

$$\frac{\partial \tilde{\vec{e_r}}}{\partial r} = 0, \qquad \frac{\partial \tilde{\vec{e_\theta}}}{\partial r} = 0$$

Key result:

$$\frac{\partial \tilde{\vec{e_r}}}{\partial \theta} = \tilde{\vec{e_\theta}}, \qquad \frac{\partial \tilde{\vec{e_\theta}}}{\partial \theta} = -\tilde{\vec{e_r}}$$

The basis vectors rotate as θ changes.

Now if you think about this, to find out the directions of the basis vectors, what we do is following the directions of the coordinate lines and in the case of polar coordinates, for the $\tilde{\vec{e_\theta}}$ basis vector, since the coordinate lines are actually curved, we need to pick the straightest possible direction when moving along it, which results in being the vector tangent to the curve.

This reasoning gives us a hint for the next important reinterpretation of basis vectors, as represented by a partial derivative of the position vector R

$$\vec{e_x} \equiv \frac{\partial \vec{R}}{\partial x} \tag{1}$$

If you carefully think about it, it might look weird at first, but it will eventually make sense. Think about **R** as a function of the coordinates x and y for example, and then consider adding a small value Δx to the x coordinate. The difference between these two position vectors results in a vector that is parallel to the x coordinate axis. So this resultant vector has the direction of the basis vector $\vec{e_x}$.
This observation is interesting, but we need to solve tha fact that the magnitude of this vector is varying , depending on the value of the Δx used.
This can be solved by dividing the resultant vector by the quantity Δx. By doing so, we obtain a vector pointing in the x-basis vector direction and magnitude 1...now we can proceed further by taking the limit as Δx approaches to 0:

$$\frac{\partial \vec{R}}{\partial x} = \lim_{\Delta x \to 0} \frac{\vec{R}(x + \Delta x, y) - \vec{R}(x, y)}{\Delta x} \tag{2}$$

Since this derivative always has a length of one and points the x-axis, we can regard it as the basis vector:

$$\vec{e_x} \equiv \frac{\partial \vec{R}}{\partial x} \tag{3}$$

This reasoning can be easily applied to the other direction, but generalized further to any coordinate system.
The neat thing about this construction is that you can do it for any position vector, and you will still get your basis vectors. Fundamentally, this trick work because, in obtaining a slightly off position vector, a small change in coordinate is added in one coordinate only while keeping other coordinates constant. Then, we divide the resultant vector (obtained from the difference between the original position vector and the new slightly off one) by that small change to normalize the vector.

Now you should remember when we discussed, in the other notes for "Tensors for Beginners" about the forward and backward transformations when we move from one basis vector coordinate system to another.
Turns out that those transformation rules can be obtained easily once we introduced this above concept about basis vectors being partial derivatives of the position vector with respect to the coordinate variables.
And also, that these transformation matrices have a proper name: **the Jacobian matrices**

Let's say we want to find the forward transformation matrix (the Jacobian) matric when moving from cartesian to polar coordinates. This would be just a matter of applying the chain rule:

$$\tilde{\vec{e_r}} = \frac{\partial \vec{R}}{\partial r} = \frac{\partial \vec{R}}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial \vec{R}}{\partial y}\frac{\partial y}{\partial r} = \frac{\partial x}{\partial r}\vec{e_x} + \frac{\partial y}{\partial r}\vec{e_y}$$
$$\tilde{\vec{e_\theta}} = \frac{\partial \vec{R}}{\partial \theta} = \frac{\partial \vec{R}}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial \vec{R}}{\partial y}\frac{\partial y}{\partial \theta} = \frac{\partial x}{\partial \theta}\vec{e_x} + \frac{\partial y}{\partial \theta}\vec{e_y}$$

We know the transformation rules between cartesian and polar coordinates:

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

so it's a matter of taking the derivatives to find that the Jacobian matrix, or forward matrix is:

$$\frac{\partial \vec{R}}{\partial r} = \frac{\partial \vec{R}}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial \vec{R}}{\partial y}\frac{\partial y}{\partial r} = cos\theta\frac{\partial \vec{R}}{\partial x} + sin\theta\frac{\partial \vec{R}}{\partial y}$$
$$\frac{\partial \vec{R}}{\partial \theta} = \frac{\partial \vec{R}}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial \vec{R}}{\partial y}\frac{\partial y}{\partial \theta} = -r\sin\theta\frac{\partial \vec{R}}{\partial x} + r\cos\theta\frac{\partial \vec{R}}{\partial y}$$

So:

$$F = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix} \tag{4}$$

And you can tackle, with the same derivation logic, the inverse Jacobian matrix, of backward transformation matrix, to obviously find out that the multiplication of the two yields the identity matrix.

Now, following a more tensorial approach, we can also re-write these transformation rules as:

$$\frac{\partial \vec{R}}{\partial p^i} = \frac{\partial \vec{R}}{\partial c^j}\frac{\partial c^j}{\partial p^i}$$
$$\frac{\partial \vec{R}}{\partial c^i} = \frac{\partial \vec{R}}{\partial p^j}\frac{\partial p^j}{\partial c^i}$$

3 Vectors as derivatives and their transformation fules

Allright we saw that basis vectors can be re-interpreted as partial derivative of the position vector along the coordinate axis.
We can sort of extend this logic to the vectors themselves. We saw that figuring out vector components in a particular basis is kind of easy to do for individual vectors, but now we're going do it for vector fields (i.e. a vector for every point in space), **but for now, we're going to deal with a slightly different type of vector field, and that's a vector field along a curve, so basically we're defining a vector at every point along a curve.**

The way we imagine vector fields along a curve is considering the set of tangent vectors to the curves. This gives us a vector field defined at every point on the curve.

A curve can be tought as a function that takes some input parameter λ and outputs the position vector $\mathbf{R}$ that traces our curve in the space.
If we want to have a vector tangent to this curve, we just need to consider taking the derivative of the position vector with respect to its parameter:

$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} \tag{5}$$

Now you may think that finding out the components of this vector in a specific basis can be difficult but it is really not, if we make use of the multivariable chain rule and expand it out in the coodinates we're working in.
For example, in the cartesian coordinates c^i and using the Einstein summation convention notation:

$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} = \frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial c^i} \tag{6}$$

At the end of the day, considering doing it in cartesian $\mathbb{R}^{\nvDash}$ for simplicity, these are equivalent (individual vector on the top, and vector field along a curve on the bottom):

$$\vec{v} = v^1\vec{e_1} + v^2\vec{e_2}$$
$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} = \frac{\mathrm{d}x}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial x} + \frac{\mathrm{d}y}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial y}$$

The same can be applied to whatever coordinate system we're working in, so to give you an example, let's say we have the circular curve of radius 2, centered at the origin in $\mathbb{R}^{\nvDash}$, defined by:

$$\vec{R}(\lambda) = (x(\lambda), y(\lambda)) = (2\cos\lambda, 2\sin\lambda) \tag{7}$$

The vector field along the curve is made up of the tangent vectors along the curve, that can be expanded as defined:

$$\begin{aligned}\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} &= \frac{\mathrm{d}x}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial x} + \frac{\mathrm{d}y}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial y}\\ &= -2\sin\lambda\frac{\partial\vec{R}}{\partial x} + 2\cos\lambda\frac{\partial\vec{R}}{\partial y}\end{aligned}$$

The components can also be written, for the cartesian basis, as a vector column:

$$\begin{bmatrix}-2\sin\lambda\\ 2\cos\lambda\end{bmatrix}_{\frac{\partial}{\partial c^i}} \tag{8}$$

We can do the same in polar coordinates r, θ, where the same curve is defined as:

$$\vec{R}(\lambda) = (r(\lambda), \theta(\lambda)) = (2, \lambda) \tag{9}$$

$$\begin{aligned}\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} &= \frac{\mathrm{d}r}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial r} + \frac{\mathrm{d}\theta}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial\theta}\\ &= 0\frac{\partial\vec{R}}{\partial r} + 1\frac{\partial\vec{R}}{\partial\theta}\end{aligned}$$

So:

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}_{\frac{\partial}{\partial p^i}} \tag{10}$$

Now, as you can already imagine, like we saw that for individual vectors, the components are contravariant (i.e. they transform the opposite way of the basis vectors), we're going to simply show that this is valid indeed in this vector fields case.

We already saw in the previous section, the transformation rules of the basis vectors using the Jacobain and Inverse Jacobian:

$$\boxed{\begin{aligned} &\frac{\partial \vec{R}}{\partial p^i} = \frac{\partial \vec{R}}{\partial c^j}\frac{\partial c^j}{\partial p^i} && \text{Forward Transform} \\ &\frac{\partial \vec{R}}{\partial c^i} = \frac{\partial \vec{R}}{\partial p^j}\frac{\partial p^j}{\partial c^i} && \text{Backward Transform} \end{aligned}} \tag{11a, 11b}$$

Now we can just use these and replace the basis vectors with their new basis transformation rule:

$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} = \frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\frac{\partial \vec{R}}{\partial c^i} = \frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\left(\frac{\partial \vec{R}}{\partial p^j}\frac{\partial p^j}{\partial c^i}\right) = \left(\frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\frac{\partial p^j}{\partial c^i}\right)\frac{\partial \vec{R}}{\partial p^j} \tag{12}$$

This is a linear combination of the new basis vector p^j, and since we know that we can write:

$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} = \frac{\mathrm{d}p^i}{\mathrm{d}\lambda}\frac{\partial \vec{R}}{\partial p^i} \tag{13}$$

This means that:

$$\frac{\mathrm{d}p^i}{\mathrm{d}\lambda} = \frac{\mathrm{d}c^j}{\mathrm{d}\lambda}\frac{\partial p^i}{\partial c^j} \tag{14}$$

Which shows that the vector field components transform using the backward transformation (inverse Jacobian)

We can actually test this out in the previous example right? So we have the components being, in both coordinate systems:

$$\begin{bmatrix} -2\sin\lambda \\ 2\cos\lambda \end{bmatrix}_{\frac{\partial}{\partial c^i}} \qquad \begin{bmatrix} 0 \\ 1 \end{bmatrix}_{\frac{\partial}{\partial p^i}}$$

And we also know the Inverse Jacobian in the case of polar-to-cartesian coordinates, being:

$$B = J^{-1} = \begin{bmatrix} x/r & y/r \\ -y/r^2 & x/r^2 \end{bmatrix} = \begin{bmatrix} \cos\lambda & \sin\lambda \\ -1/2\sin\lambda & 1/2\cos\lambda \end{bmatrix} \tag{15}$$

Let's check that:

$$
\begin{aligned}
B \begin{bmatrix} -2\sin\lambda \\ 2\cos\lambda \end{bmatrix}_{\frac{\partial}{\partial c^i}} &= \begin{bmatrix} \cos\lambda & \sin\lambda \\ -1/2\sin\lambda & 1/2\cos\lambda \end{bmatrix} \begin{bmatrix} -2\sin\lambda \\ 2\cos\lambda \end{bmatrix}_{\frac{\partial}{\partial c^i}} \\
&= \begin{bmatrix} -\cos\lambda 2\sin\lambda + \sin\lambda 2\cos\lambda \\ -1/2\sin\lambda(-2\sin\lambda) + 1/2\cos\lambda 2\cos\lambda \end{bmatrix}_{\frac{\partial}{\partial p^i}} \\
&= \begin{bmatrix} 0 \\ 1 \end{bmatrix}_{\frac{\partial}{\partial p^i}}
\end{aligned}
$$

Same can be checked on the opposite direction applying the forward/jacobian transformation.

One last thing to mention before terminating this section is in terms of notation: you notice that the position vector appears in a lot of them, so what we're going to do from here onward, is get rid of the position vector symbol.
This might seem like a weird choice of notation, but from now on, for example, the $\mathbb{R}^2$ cartesian basis vectors will be written as:

$$
\begin{aligned}
\vec{e_x} &= \frac{\partial}{\partial x} \\
\vec{e_y} &= \frac{\partial}{\partial y}
\end{aligned}
$$

where basically we consider basis vectors these partial derivative operators.
If you think about this, it's not very weird at the end, as derivative operators are linear, as they can be scaled and added like vectors do, so they have both a magnitude and a direction. They also obey the forward and backward transformation rules that we would expect vectors to obey.

The main logical reason to get rid of the position vector symbol is that, the position vector definition requires an origin to be set, and later on when we're going to deal with curved surfaces and manifolds, it's not easy to define an origin point.
So, using a position vector without an origin does not make a lot of sense, but we're still going to be able to use vectors and vector fields as long as we have these derivative operators to work with.

The whole main idea is the following:
Basis transforms with the Jacobian;
Components transform with the inverse Jacobian;
Their product gives the identity;
Therefore the geometric vector is coordinate-invariant.

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} = \frac{\mathrm{d}p^i}{\mathrm{d}\lambda}\frac{\partial}{\partial p^i} = \frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\frac{\partial}{\partial c^i} \qquad \text{Total derivative along a curve (chain rule)} \tag{16a}$$

$$\frac{\partial}{\partial p^i} = \frac{\partial}{\partial c^j}\frac{\partial c^j}{\partial p^i} \qquad \text{Coordinate basis transformation} \tag{16b}$$

$$\frac{\partial}{\partial c^i} = \frac{\partial}{\partial p^j}\frac{\partial p^j}{\partial c^i} \qquad \text{Inverse coordinate basis transformation} \tag{16c}$$

$$\frac{\mathrm{d}p^i}{\mathrm{d}\lambda} = \frac{\mathrm{d}c^j}{\mathrm{d}\lambda}\frac{\partial p^i}{\partial c^j} \qquad \text{Contravariant component transformation} \tag{16d}$$

$$\frac{\mathrm{d}c^i}{\mathrm{d}\lambda} = \frac{\mathrm{d}p^j}{\mathrm{d}\lambda}\frac{\partial c^i}{\partial p^j} \qquad \text{Inverse component transformation} \tag{16e}$$

3.1 Hold on - Why are derivative operators vectors?

Let's here make some clarity about what we've discussed so far in the previous two sections. We brought up this new way of thinking about basis vectors, where we went from considering basis vectors as derivative of the position vector wrt the coordinate variables, to them being defined by the partial derivative operators themselves.
Let's start by saying that this way of thinking will be appreciated more when we'll start thinking anout curved (differential) geometry, but we chose to introduce it right now, even though we're still working in flat space, to already start using these ideas and notations beforehand.

We just a bit ahead of ourselves here, to explain the main reason of already working with this, so you get an idea of what we're going to deal with, and then we'll get back to our flat space for now.

We saw that if we consider the difference between two position vectors, a small distance h apart from each other in the direction of one coordinate axis, divide this by this distance h, and consider the limit when h approaches 0, gives us as result the basis vector of magnitude 1 in the considered axis direction. We we said that this position vector derivative along the axis-variable and the unit vector along this axis are equivalent.

We also shown that, for tangent vectors along a curve, parametrized by the paramter λ, considering the position vector at a certain position λ and another one a small step h away, we can compute the tangent vector to this curve at position λ, by taking the difference and consider the limit when h approaches 0.

And we can expand the tangent vector out as a linear combination of the basis vectors using the chain rule. Now all of this, has been discussed working on the flap 2D plane.

Let's try to move further and consider this in a curved space, for example the surface of the earth, to see how the situation is different.

If we consider a point moving on the surface of the earth from one position to another, and we want to compute its velocity, we need to find the tangent vector to the curve he moves along. So what do we do? We can take the derivative of the position vector wrt to the parameter we use to define this curve.
Okay, so how do we define the position vector? We need to define an origin, right? We can think of putting this origin at the center of the sphere (earth)....but this is a bit of a different situation than before when working in the flat space?

Why? Well first because the origin does not live on the earth's 2D surface, and second because the velocity vector itself, also does not live on the earth's 2D surface, since it extends to the outside space.
They both are external to the surface we're studying, and indeed this is referred to "extrinsic geometry"

Now, this may not seem a problem, if we consider the 3D surface of a globe, from an external perspective of a 3D space.
But what if instead of this situation, we're forced to look at the surface of the earth on a map?

How do we define position vectors here? There's not really a convenient outside place where we can put the origin here, there's no external place for us to use. You might think that you can potentially use any place on the map to put the origin, choosing a random placefor it and drawing a straight line representing the position vector.

There's a problem with this approach though, vectors are supposed to be straight, but the path from the origin to the tip of the vector is not really straight, because the surface in reality is not flat.

The method of studying curved surfaces where we're not allowed to leave the surface is called intrinsic geometry.

When we'll study General relativity, we're going to deal with a 4-dimensional curved spacetime geometry, the situation we're running on is exactly this. Spacetime is curved so we cannot really run straight lines, hence we cannot draw position vectors.

And we cannot pick an origin point outside of this 4-dim space, because that would mean picking a point outside universe, and that's completely meaningless.

So the summary is that, we're forced to use intrinsic geometry in General Relativity, so how are we supposed to define things like velocity, acceleration etc if we're not allowed to use position vectors?

The solution is, to realize that, while we cannot draw straight lines on a curved surface, we can still draw a curved path, and on a map, we can use grid lines, so we can still define derivatives wrt these curves. So we can use these derivative operators to talk about different directions on the surface.
The partial derivative wrt to x for example, points to the x direction, the partial wrt to y points to the y direction and so on.

If we have some path traveling around in our curved space, we can still consider the direction that the path is pointing to, using the derivative wrt the curve parameter used, and we break this direction up in terms of our x, y components, using the multi-variable chain rule as we saw.

Formally, our new notation with basis vectors defined as partial derivative operators along the axis directions, form a vector space called the "Tangent Vector Space" T_pM, which is the vector space of derivatives at some point p on a surface M

References

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